



Mathematics Extension 1
Year 11 Assessment Task 2
Week 4 Term 3 2021

Name:

Teacher:

Thursday 5th August 2021

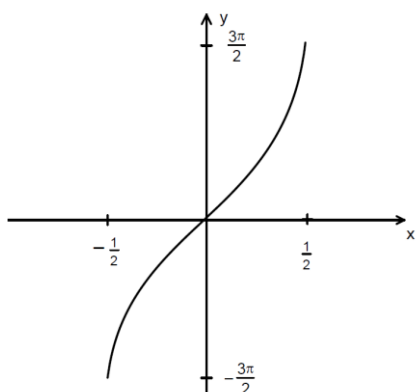
General Instructions

- All answers are to be written on your own lined paper so they can be scanned and uploaded as a single PDF file to your teacher's google classroom. Make sure that you write your name on each of the lined pages used.
- **Start each question on a new page.**
- Calculators approved by NESA may be used.
- Write using black pen.
- In Questions 1– 5, show relevant mathematical reasoning and/or calculations. Marks may be deducted for insufficient or illegible work.
- For any multiple-choice question, **write the letter** that corresponds to your answer on your writing paper.
- Time Allowed: **55 minutes inclusive of task downloading then answering on your own lined paper, scanning and uploading a single PDF file for your responses to your teacher's google classroom.**
- **Total marks – 32**

Setter: YUS

Question 1 (6 marks)

- (a) Which of the alternatives could represent the inverse trigonometric function drawn below?



- A. $y = \frac{1}{3} \sin^{-1} \frac{x}{2}$
- B. $y = \frac{1}{3} \sin^{-1} 2x$
- C. $y = 3 \sin^{-1} \frac{x}{2}$
- D. $y = 3 \sin^{-1} 2x$

- (b) If $\log_5 8 = a$, prove that $\log_{10} 2 = \frac{a}{a+3}$. 2

- (c) An inverted cone of radius $r = 40$ cm and depth $h = 70$ cm is filled with water. The water flows out of a hole in the apex of the cone at a constant rate of $30\pi \text{ cm}^3/\text{minute}$.

- (i) Show that the volume V of the cone is given by $V = \frac{16\pi h^3}{147}$. 1

- (ii) Find the rate at which the water level is decreasing when the depth of water is 10cm. 2

End of Question 1

Question 2 (6 marks) Start a new page.

A baked cake cools at a rate proportional to the difference between the cake temperature (T) and the room temperature (20°C). Initially the cake is 80°C and after 10 minutes the cake is 50°C .

- (i) Show that $T = 20 + Ae^{kt}$ satisfies the cooling equation $\frac{dT}{dt} = k(T - 20)$. 1
- (ii) Show that $A = 60$. 1
- (iii) Find the exact value of k and hence find when the cake temperature reaches 35°C . 2
- (iv) Find the limiting temperature of the cake. 1
- (v) What is the rate of cooling after exactly 10 minutes? 1

End of Question 2

Question 3 (7 marks) Start a new page.

- (a) A driver takes 3 hours to travel the distance between two points P and Q on a country road. At time t hours after passing P , the driver's speed v km/h is given by $v = 70 + 40e^{-t}$.
 - (i) Calculate the speed when the driver passes point Q , to the 2 significant figures. 1
 - (ii) Write the acceleration in terms of t . 1
 - (iii) Sketch the velocity-time curve and comment on the velocity for large values of t . 2
- (b) Find the value of $\tan\left(\frac{7\pi}{12}\right)$ as a simplified surd with a rational denominator. 3

End of Question 3

Question 4 (6 marks) Start a new page.

- (a) Using the t -substitutions, or otherwise, prove the identity

3

$$\frac{\tan 2\theta - \tan \theta}{\tan 2\theta + \cot \theta} = \tan^2 \theta$$

- (b)

$$\text{For the function } f(x) = \begin{cases} \frac{1}{x} & \text{for } 0 < x < 2 \\ \frac{1}{2} & \text{for } x = 2 \\ 1 - \frac{1}{4}x & \text{for } x > 2 \end{cases}$$

- (i) Show that the curve is continuous at $x = 2$.
- (ii) Determine whether the function is differentiable at $x = 2$ and if so, state the value of $f'(2)$. Explain your reasoning carefully.

1

2

End of Question 4

Question 5 (7 marks) Start a new page.

- (a) Determine where the curve $y = \sin 5x + \sin x$ cuts the x -axis for $0 \leq x \leq \frac{\pi}{2}$.

2

- (b) Consider the function $y = \tan^{-1}(\tan x)$.

- (i) State its domain and whether it is even, odd or neither

2

- (ii) Simplify $\tan^{-1}(\tan x)$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

1

- (iii) Use the above information and a table of values if necessary to sketch the function for $-\frac{\pi}{2} \leq x \leq \frac{3\pi}{2}$. You may assume $\tan x = \tan(x + \pi)$.

2

End of paper



Suggested Solution (s)

Comments

Question 1

a) $-\frac{1}{2} \leq x \leq \frac{1}{2}$

$-1 \leq 2x \leq 1$

& $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$

$-\frac{\pi}{2} \leq \frac{y}{3} \leq \frac{\pi}{2}$

So $\frac{y}{3} = \sin^{-1} 2x$

(ie) $y = 3\sin^{-1} 2x$

b) $\log_5 8 = a \Rightarrow \log_5 2^3 = a$

(ie) $\log_5 2 = \frac{a}{3}$

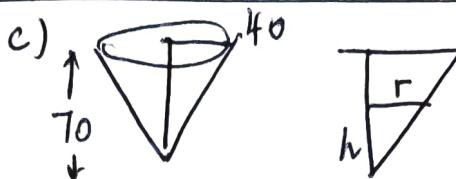
Now $\log_{10} 2 = \frac{\log_5 2}{\log_5 10}$

$= \frac{\log_5 2}{\log_5 (2 \times 5)}$
 $= \frac{\log_5 2}{\log_5 2 + \log_5 5}$ or $\checkmark$

$\therefore = \frac{a/3}{a/3 + 1}$ using $(*)$

$= \frac{a/3}{a/3 + 3/3}$

$= \frac{a}{a+3}$



i) $V = \frac{1}{3} \pi r^2 h$ but $\frac{r}{h} = \frac{40}{70}$

$r = \frac{4h}{7}$

$\therefore V = \frac{1}{3} \pi \left(\frac{4h}{7} \right)^2 h$ $\checkmark$
 $= \frac{16\pi h^3}{147}$

ii) Now $\frac{dV}{dh} = \frac{48\pi h^2}{147}$

$\frac{dh}{dt} = \frac{dh}{dV} \cdot \frac{dV}{dt}$

$= \frac{147}{48\pi h^2} \times -30\pi$ $\checkmark$

Now $h = 10$

$\frac{dh}{dt} = \frac{147 \times -30\pi}{48\pi (10)^2}$

$= \frac{-4410}{4800}$ $\checkmark$

$= -0.91875 \text{ cm/min.}$

or $-\frac{147}{160} \text{ cm/min.}$

$\therefore$ The depth of water is decreasing at 0.92 cm/min.



Year 11

Mathematics Extension 1

SOLUTIONS

2

Suggested Solution (s)

Comments

Question 2

(i) $T = 20 + Ae^{kt}$

$$\begin{aligned} \text{LHS} = \frac{dT}{dt} &= Ae^{kt} \times k \\ &= Ake^{kt} \end{aligned}$$

but $Ae^{kt} = T - 20$ ✓

$$\begin{aligned} \therefore \frac{dT}{dt} &= k(T - 20) \\ &= \text{RHS} \end{aligned}$$

(ii) When $t = 0$ $T = 80$

ie) $80 = 20 + Ae^0$ ✓

$A = 60$

(iii) When $t = 10$, $T = 50$

ie) $50 = 20 + 60e^{10k}$

$$\frac{30}{60} = e^{10k}$$

$\ln \frac{1}{2} = 10k$

$-\ln 2 = 10k$

$$\therefore k = -\frac{\ln 2}{10}$$
 ✓

When $T = 35$, $t = ?$

ie) $35 = 20 + 60e^{kt}$ where $k = \frac{-\ln 2}{10}$

$$\frac{15}{60} = e^{kt}$$

$\ln \frac{1}{4} = kt$

$$t = \frac{\ln \frac{1}{4}}{k}$$

$t = 20 \text{ min}$ ✓

iv) As $t \rightarrow \infty$ $60e^{kt} \rightarrow 0$

$\therefore T \rightarrow 20$

The limiting temperature is 20° ✓

v) $\frac{dT}{dt} = k(T - 20)$

$= k(50 - 20)$

where $k = -\frac{\ln 2}{10}$

$= -2.079441$

The rate of cooling

is 2.079°C/min ✓



Year 11

Mathematics Extension 1

SOLUTIONS

3

Suggested Solution (s)

Comments

Question 3

a) $V = 70 + 40e^{-t}$

i) For Q, $t = 3$

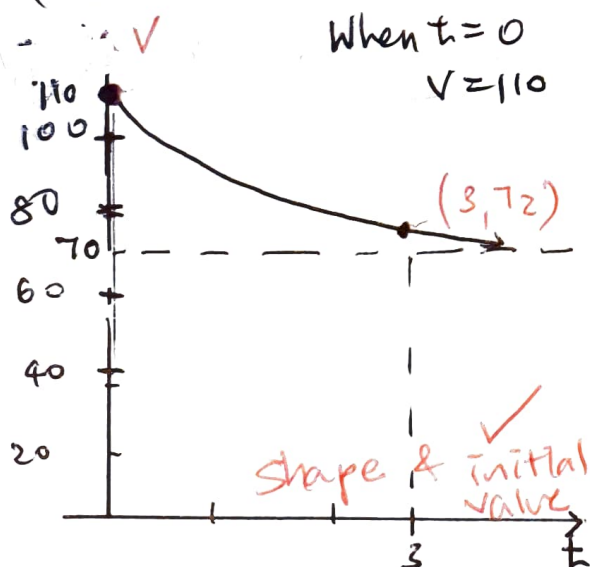
$\therefore V = 70 + 40e^{-3}$

$= 71.99$

$\approx 72 \text{ km/h}$ ✓

ii) $a = -40e^{-t}$ ✓

(iii)



As $t \rightarrow \infty$, $40e^{-t} \rightarrow 0$

$\therefore V \rightarrow 70$

as t increases the velocity approaches 70 km/h ✓

(b) $\tan \frac{7\pi}{12} = \tan \left(\frac{3\pi}{12} + \frac{4\pi}{12} \right)$

ie) $\tan \left(\frac{\pi}{4} + \frac{\pi}{3} \right)$

$= \frac{\tan \frac{\pi}{4} + \tan \frac{\pi}{3}}{1 - \tan \frac{\pi}{4} \tan \frac{\pi}{3}}$ ✓

$= \frac{1 + \sqrt{3}}{1 - \sqrt{3}}$ ✓

$= \frac{(1 + \sqrt{3})(1 + \sqrt{3})}{(1 - \sqrt{3})(1 + \sqrt{3})}$

$= \frac{1 + 2\sqrt{3} + 3}{-2}$

$= -2 - \sqrt{3}$ ✓



Suggested Solution (s)

Comments

Question 4Method I

Let $\tan \theta = t$

$$\text{LHS} = \frac{\frac{2t}{1-t^2} - t}{\frac{2t}{1-t^2} + \frac{1}{t}}$$

RHS = t^2

$$= \frac{2t - t(1-t^2)}{1-t^2} \div \frac{2t^2 + (1-t^2)}{t(1-t^2)} \quad \checkmark$$

$$= \frac{2t - t + t^3}{(1-t^2)} \times \frac{t(1-t^2)}{2t^2 + 1 - t^2} \quad \checkmark$$

$$= (t^3 + t) \times \frac{t}{t^2 + 1} \quad \checkmark$$

$$= t(t^2 + 1) \times \frac{t}{(t^2 + 1)} \quad \checkmark$$

$$= t^2$$

$$= \text{RHS}$$

Method II

$$\frac{\frac{2\tan \theta}{1-\tan^2 \theta} - \frac{\tan \theta}{1}}{\frac{2\tan \theta}{1-\tan^2 \theta} + \frac{1}{\tan \theta}}$$

$$= \frac{(\tan \theta + \tan^3 \theta) / (1 - \tan^2 \theta)}{(\tan^2 \theta + 1) / \tan \theta (1 - \tan^2 \theta)} \quad \checkmark$$

$$= \frac{\tan \theta (1 + \tan^2 \theta)}{(1 - \tan^2 \theta)} \times \frac{\tan (1 - \tan^2 \theta)}{(\tan^2 \theta + 1)} \quad \checkmark$$

$$= \tan^2 \theta$$



2021 Year 11 Mathematics Extension 1 Task 2 SOLUTIONS

Suggested Solution (s)

Comments

Question 4 (b)

(i) Given $f(2) = \frac{1}{2}$, we require $\lim_{x \rightarrow 2^-} f(x) = f(2) = \lim_{x \rightarrow 2^+} f(x)$.

Now, $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{1}{x}$ or $\frac{1}{2}$ & $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (1 - \frac{1}{4}x)$
 $= 1 - \frac{1}{4} \times 2$ or $\frac{1}{2}$.

As $\lim_{x \rightarrow 2^-} f(x) = f(2)$ or $\frac{1}{2} = \lim_{x \rightarrow 2^+} f(x)$, f is cti at $x=2$

(ii) We require $\lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h}$ & $\lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h}$ to

exist with $\lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h}$

Now, $\lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h}$

$$= \lim_{h \rightarrow 0^-} \frac{\frac{1}{2+h} - \frac{1}{2}}{h}$$

$$= \lim_{h \rightarrow 0^-} \left(\frac{2 - (2+h)}{2(2+h)} \times \frac{1}{h} \right)$$

$$= \lim_{h \rightarrow 0^-} \frac{-h}{2h(2+h)}$$

$$= \lim_{h \rightarrow 0^-} \frac{-1}{2(2+h)}$$

$$= -\frac{1}{4}$$

& $\lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h}$

$$= \lim_{h \rightarrow 0^+} \frac{1 - \frac{1}{4}(2+h) - \frac{1}{2}}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{1 - \frac{1}{2} - \frac{1}{4}h - \frac{1}{2}}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{-\frac{1}{4}h}{h}$$

$$= \lim_{h \rightarrow 0^+} -\frac{1}{4}$$

$$= -\frac{1}{4}$$

As $\lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h}$, conclude

that the function is differentiable at $x=2$ with $f'(2) = -\frac{1}{4}$.

2 marks
for
argument.



Suggested Solution (s)

Comments

Question 5

a) $y = \sin 5x + \sin x$

To cut x -axis

$$\sin 5x + \sin x = 0$$

$$\sin(3x+2x) + \sin(3x-x) = 0$$

ie) $2 \sin 3x \cos 2x = 0$ ✓

So $\sin 3x = 0$ or $\cos 2x = 0$

$$3x = 0, \pi, 2\pi, 3\pi, \dots$$

or $2x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$ ✓

∴ $x = 0, \frac{\pi}{3}$ or $x = \frac{\pi}{4}$ ✓

(b) For $y = \tan^{-1}(\tan x)$

(i) Domain: all real x

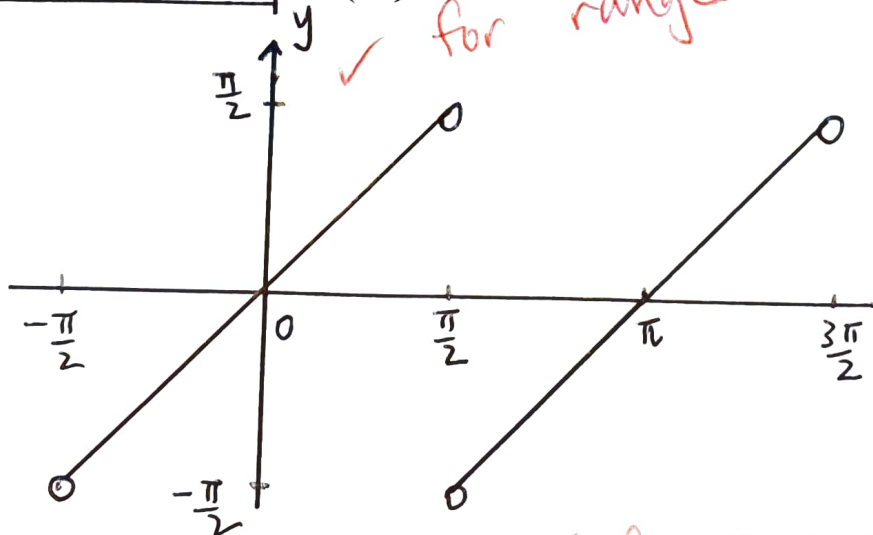
except $x = \frac{(2n+1)\pi}{2}$ ✓

where n is an integer

& Odd function ✓

(ii) $\tan^{-1} \tan x = x$ ✓

(iii)



✓ for range

✓ for 2 straight line

□ Carried error for 'open circle' from (i)